MILITARY SPECIFICATION

TELESCOPE: XM119

1. SCOPE

1.1 This specification covers Telescope XM119. An articulated joint permits the objective tube to elevate coaxially with the weapon while the eyepiece remains stationary (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F8624500	Telescope, XM119

FSC 1240

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DRAWINGS (Continued)

U.S. Army Munitions Command

F8565556	Pressure Tester, Two Station
IEL 8624500	List of Inspection Documents for Telescope: XM 119

PACKAGING DATA SHEET

MIL-P-14232/P8624500	Packaging of Telescope: XM119

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with Drawing F8624500 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926.

MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connection

3.3 Environmental.

3.3.1 Storage temperature.- The telescope shall show no evidence of physical failure such as cement separation, glass breakage, or casting failure after having been exposed and thermally stabilized at temperatures of -80° and +160° Fahrenheit (F).

3.3.2 Extreme operating temperature.- The telescope shall meet the applicable requirements of 3.16 while exposed and thermally stabilized at -40° and +150°F.

3.3.3 Vibration.- The telescope shall show no evidence of physical failure due to being vibrated at a constant frequency of 30 cycles per second (cps) with an amplitute of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds.

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3.4 Cleanliness.- The telescope shall meet the cleanliness requirements of MIL-O-13830 when viewed through the eyepiece. The telescope shall show no more than five particles of dirt 0.2 mil maximum real diameter in the outer 35 degrees, and four particles of dirt 0.05 mil maximum real diameter in the central 25 degrees of the apparent field. There shall be no other dirt which shall adversely affect observation and image quality.

3.5 Sealing.- The telescope shall show no evidence of leakage when subjected to an internal pressure of 5 ± 0.10 pound per square inch (psig).

3.6 Orientation.- With the geometric axis of the objective tube (front tube) in a horizontal plane as established by its spherical mounting surfaces, the locating key at the six o'clock position; the center of the eyepiece hanger bolt hole shall be positioned 2.822 ± 0.010 inches below the plane of the geometric axis of the objective tube.

3.7 Collimation.- With the geometric axis of the missile reticle in coincidence with the geometric axis of the telescope the following requirements shall be met.

3.7.1 Reticle position.

3.7.1.1 Conventional reticle.- The boresight cross of the conventional reticle shall be 26.6 mils ± 0.25 mil above the geometric axis of the telescope and be in coincidence with a plumb line passing through the geometric axis.

3.7.1.2 Missile reticle.- The boresight collimating circle of the missile reticle shall be set at the 12 o'clock position.

3.7.2 Missile reticle movement.- From the position established in 3.7 rotation of the vertical boresight knob shall move the line of sight 3.5 mils in depression and 3.5 mils in elevation; rotation of the horizontal boresight knob shall move the line of sight 3.5 mils to the left and 3.5 mils to the right.

3.7.3 Conventional reticle movement.- From the position established in 3.7.1 rotation of the vertical boresight knob shall move the line of sight 6 mils in depression and 9 mils in elevation; rotation of the horizontal boresight knob shall move the line of sight 7.5 mils to the left and 7.5 mils to the right.

3.7.4 Backlash.- Backlash in any boresight mechanism shall not exceed 0.1 mil.

3.8 Reticle wander.- The image of the reticle boresight mark shall not deviate by more than one degree from a fixed reference in an auxiliary telescope secured to the eyepiece of the telescope when the telescope is articulated through its full range (from stop to stop).

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3.9 Reticle lean.- The angular rotation of the missile reticle with respect to a reference in an auxiliary telescope secured to the eyepiece shall not exceed one degree when the telescope is articulated through its full range.

3.10 Image tilt.- The image of a vertical target shall be vertical within one degree when the telescope is positioned as in 3.6. (One degree tilt is equivalent to the angle subtended by a 2.5 mil block at the edge of the field when measured across the full field of view).

3.11 Resolution.- The resolution on the optical axis shall be 6.0 seconds of arc or less.

3.12 Eyepiece focus.- Eyepiece focus shall be adjustable over the range of plus and minus 4 diopters as read on the eyepiece scale. The zero setting of the diopter scale shall indicate zero diopter within 0.25 diopter when the image of an infinity target is in sharp focus.

3.13 Reticle accuracy.- The angular values for reticle graduation between the zero and maximum range marks shall be as indicated on the reticle drawing within 0.2 mil.

3.14 Parallax.- The total parallax between either reticle and the target image at the center of the field of view shall not exceed 0.1 mil, when viewing a target between 600 meters and infinity, with the parallax adjustment set at mid-point of total travel.

3.15 Parallelism of reticle and image.- With the telescope positioned in accordance with 3.6 the following requirements shall be met.

3.15.1 Missile reticle.- The vertical reticle line of the missile reticle shall be parallel to the image of a plumb line within 30 minutes of arc.

3.15.2 Conventional reticle.- The representative vertical reticle line (total area of dashes and spaces) shall be parallel to the image of a plumb line within 30 minutes of arc.

3.15.3 Variation of settings.- The angular difference between the missile reticle and the conventional reticle shall not exceed 30 minutes of arc.

3.16 Operability.

3.16.1 Diopter adjustment torque.- The torque required to rotate the diopter adjustment shall be not less than 10 nor more than 20 inch pounds at standard ambient temperature (plus 60 degrees to 90°F) and shall not exceed 25 inch pounds at -40° and +150°F.

3.16.2 Filter transfer lever.- The filter transfer lever shall be operable and shall firmly seat against the stops at -40° and +150°F.

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3.16.3 Articulation torque.- With the telescope on its right or left side, unrestricted, the torque required to rotate the objective tube about the articulated joint through full movement shall not exceed 50 foot pounds at a standard ambient temperature between 60° and 90°F, and 75 foot pounds at -40° and +150°F.

3.16.4 Boresight knobs.- The force required to disengage the boresight knobs from their normal position shall be 4 plus or minus 2 pounds at standard ambient temperature and shall be 6 plus or minus 2 pounds at -40° and +150°F.

3.16.4.1 Rotation.- The torque required to rotate the disengage boresight knobs shall be 3 ± 1 inch pounds at standard ambient temperature, and shall be 5 plus or minus 2 pounds at -40° and +150°F.

3.17 Lamp housing assemblies.- The lamp housing assemblies shall be mechanically operable at standard ambient temperature and at extreme temperatures of 3.3.2 to facilitate replacement of reticle lamp.

3.18 Electrical.

3.18.1 Reticle illumination.- With the reticle light for each reticle energized by a 24 volt power source, the illuminated reticle shall appear clearly defined when observed in a darkened area when the filters are in the clear position.

3.18.2 Circuitry.- The circuitry of the telescope shall be in accordance with the applicable electrical schematic drawings.

3.18.3 Dielectric strength.- There shall be no puncture of insulation, no sparkover, no decrease in dielectric strength, when the insulated areas of the circuitry is subjected to an alternating current (ac) root mean square (rms) potential of 500 volts plus twice the rated voltage at a frequency of 60 cps for a period of 15 plus or minus 5 seconds.

3.18.4 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.19 Interior atmosphere.- The interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F and the internal pressure of the telescope shall be between 0.10 and 0.25 psig.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 8624500. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65 percent defective			
101.	Vibration	3.3.3	4.5.3
102.	Cleanliness	3.4	4.5.4
103.	Collimation	3.7	4.5.7.1
104.	Conventional reticle position	3.7.1.1	4.5.7.1.1
105.	Missile reticle position	3.7.1.2	4.5.7.1.2
106.	Missile reticle movement	3.7.2	4.5.7.2
107.	Conventional reticle movement	3.7.3	4.5.7.3
108.	Backlash, Missile reticle boresight mechanism	3.7.4	4.5.7.4
109.	Backlash, Conventional reticle boresight mechanism	3.7.4	4.5.7.7.5
110.	Reticle wander	3.8	4.5.7.6

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		REQUIREMENT	TEST PROCEDURE
111.	Reticle lean	3.9	4.5.7.7
112.	Image tilt	3.10	4.5.7.8
113.	Resolution	3.11	4.5.7.9
114.	Eyepiece focus	3.12	4.5.7.10
115.	Reticle accuracy	3.13	4.5.7.11
116.	Parallax	3.14	4.5.7.12
117.	Parallelism of missile reticle and image	3.15.1	4.5.7.13
118.	Parallelism of conventional reticle and image	3.15.2	4.5.7.14
119.	Variation of settings	3.15.3	4.5.7.15
120.	Diopter adjustment, torque	3.16.1	4.5.8
121.	Filter, transfer lever, operability	3.16.2	4.5.9
122.	Articulation torque	3.16.3	4.5.10
123.	Boresight knobs - force	3.16.4	4.5.11
124.	Boresight knobs - rotation (torque)	3.16.4.1	4.5.12
125.	Lamp housing assemblies, operability	3.17	4.5.13
126.	Reticle illumination	3.18.1	4.5.14
127.	Circuitry	3.18.2	4.5.15
128.	Dielectric strength	3.18.3	4.5.16
129.	Insulation resistance	3.18.4	4.5.17

MINOR: NONE

NOTE: The tests of Table I shall be conducted at standard ambient temperature (Plus 60° to 90°F).

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests.

4.3.1 General.- One telescope shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST		REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Visual-Per drawings
302.	General specification	3.2	Visual-MIL-F-13926

4.3.2 Environmental.- Three telescopes shall be selected at random as control samples from each 50 produced, or from each month's production, whichever occurs first. The control samples shall meet the following requirements and tests:

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CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Storage temperatures	3.3.1	4.5.1
304. Operability at extreme temperatures	3.3.2	4.5.2
305. Sealing	3.5	4.5.5
306. Interior atmosphere	3.19	4.5.6

4.3.3 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents, IEL-8624500.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 Inspection equipment design requirements.

4.4.2.3.1 Special testing equipment to test sealing.- The pressure testing equipment to test sealing of the telescope shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

4.4.2.3.2 Special testing equipment utilized for orientation of telescope.- The testing equipment utilized to mount and orient the telescope for the accuracy tests shall permit mounting the geometric axis of the objective tube (front tube) in a horizontal plane within ten seconds, the locating key at the six o'clock position within ten seconds, and the center of the eyepiece hanger bolt hole positioned 2.822 $\pm$ 0.010 below the plane of the objective tube. The testing equipment shall also provide for articulation of the eyepiece housing assembly to any position in elevation between its stops.

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4.4.2.3.2.1 Special testing equipment to test collimation, reticle position, missile reticle movement and conventional reticle movement.- The testing equipment for these tests shall be accurate within eighteen seconds.

4.4.2.3.2.2 Special testing equipment to test reticle wander, reticle lean and image tilt.- The testing equipment utilized for these tests shall be accurate within six minutes.

4.4.2.3.2.3 Special testing equipment to test reticle accuracy, and parallax.- The testing equipment used for these tests shall be accurate within three minutes.

4.4.2.3.2.4 Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.4.2.3.3 Special testing to test torque requirements.- A torque wrench type tester with a minimum range from zero to twenty pound inches, the accuracy of which shall be plus or minus one-quarter pound inch throughout the entire zone of excursion, and a torque wrench type tester with a minimum range from zero to seventy-five pound feet, the accuracy of which shall be plus or minus one-half pound foot throughout the entire zone of excursion, shall be utilized. The testing equipment shall be equipped with suitable adapters to accomodate testing of the diopter adjustment, objective tube and the boresight knobs.

4.5 Test methods and procedures.

4.5.1 Storage temperatures.- The telescope shall be exposed to the temperatures indicated in 3.3.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. This equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. The telescope shall be exposed to each specified temperature extreme for a minimum of 5 hours, and then permitted to return to an ambient temperature between +60° and 90°F. The telescope shall be subjected to a visual and tactile examination to determine that there is no evidence of mechanical and optical failure or deterioration. The telescope shall conform to the requirements of 3.3.1.

4.5.2 Extreme operating temperatures.- The telescope shall be exposed to the temperatures indicated in 3.3.2, utilizing testing equipment in accordance with "Test Facilities" requirement of MIL-F-13926. The equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. Temperatures shall be changed gradually to avoid thermal shock. When thermal stabilization is reached at each temperature extreme, the telescope shall be subjected to the tests specified in 4.5.7 to 4.5.12 inclusive. The indicated controls and assemblies shall conform to the operability requirements for each temperature extreme.

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4.5.3 Vibration.- The telescope shall be subjected to the vibration test utilizing standard vibrating equipment capable of meeting the conditions of 3.3.3. The accuracy of the applicable test equipment and test equipment measurements shall be in accordance with the "Test facilities" requirements of MIL-F-13926. The telescope shall be horizontally mounted on the vibrating table by means of an adapter. This adapter shall immobilize the objective tube assembly, and the offset tube assembly, so that the articulating joint assembly is immobilized in its midpoint of the articulating range. The telescope shall be vibrated in a vertical plane with the motion and frequency applied for 5 minutes ± 15 seconds. At the conclusion of this vibration the telescope shall be subjected to a visual and tactile examination for damage such as: bent or broken parts, glass chips, fractures, foreign matter, functioning of moving parts for undue irregularities of movement, friction or looseness. Any one of the above possible defects shall be cause to consider a telescope defective unless separate defects-per-hundred type sampling plans are established or approved by the Government as part of the supplier's inspection system. Telescopes which have passed the above examinations shall be subjected to the tests of 4.5.4 to 4.5.17 inclusive and shall conform to the requirements specified therein.

4.5.4 Cleanliness.- The optical surfaces shall be inspected for cleanliness in accordance with MIL-O-13830. The telescope shall conform to the cleanliness requirements of 3.4. Dirt particles which interfere with the legibility or accuracy of the reticle patterns shall be cause for rejection.

4.5.5 Sealing.- The internal pressurizing of the telescope, to test sealing, shall be performed with special testing equipment conforming to 4.4.2.3.1. The pressure shall be applied to the telescope through the valve in the offset housing. Pressurize the telescope with dry nitrogen gas to the value specified in 3.5, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the telescope for a period of 5 minutes ± 15 seconds. During this time period, the telescope eyepiece housing assembly shall be elevated through one full excursion of the articulated joint assembly, and the diopter adjustment shall be rotated one full excursion clockwise and counterclockwise. At the end of the 5 minute period, there shall be no evidence of leakage as indicated on the pressure gauge of the testing equipment, as required in 3.5. Repeat this test procedure for the reticle projector assembly, except that the pressure shall be applied to the valve on the reticle projector assembly, and while the assembly is pressurized, rotate the elevation and deflection knobs located at the bottom side of the assembly. The reticle projector assembly shall show no evidence of leakage.

4.5.6 Interior atmosphere.- Upon satisfactory completion of the sealing test the telescope shall be flushed with the dry nitrogen gas. The flushing shall be accomplished by opening the purging vent on the objective tube and allowing the pressure used for testing to flow through the telescope, until gas bled from the telescope conforms to the dew point requirements of 3.19, as measured with a dew point tester. Close the purging vent and reduce the internal pressure of the telescope to the value specified in 3.19, as measured with sensitive pressure testing equipment. The same procedure for flushing shall be applied to the reticle projector assembly by opening the purging vent on the housing assembly containing the elevation knob.

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4.5.7 Orientation.- Orientation shall be accomplished by means of special testing equipment in compliance with 4.4.2.3.2, placed on a surface plate or table that has been leveled by means of a precision (10 seconds) bench level. The end of the objective tube assembly shall be positioned toward a target for measuring the projected reticle patterns. Unless otherwise specified, optical accuracy of the telescope shall be measured on a collimator projector target set to a focus of 1200 meters ± 100 meters. The optical axis of the collimator projector shall be coincident to the geometric axis of the telescope. A reticle switch selector to select one of the two reticle patterns for projection into the optical path of the telescope, and a 28 volt dc power source to provide the power to energize the telescope will be used in conjunction with the testing equipment specified in 4.4.2.3.2 in performance of the collimation test and subsequent tests.

4.5.7.1 Collimation.- With the telescope oriented as required, actuate reticle selector switch to illuminate the missile reticle. Position the geometric axis of the missile reticle (intersecting point of the vertical and horizontal lines) to be coincident to the intersecting point of the vertical and horizontal target line image on the collimator projector target to determine conformance with 4.7. A reference mark shall be made on the dial control plate in true alignment with a reference mark on the housing which will serve as a zero reference point or midpoint position, for the missile boresight knobs.

4.5.7.1.1 Conventional reticle position.- Project the conventional reticle pattern into the optical path of the telescope by means of the reticle selector switch. Adjust the boresight knobs so that a line of sight through the geometric axis of the telescope reticle is coincient to the intersecting point of the vertical and horizontal target line image. The angular subtense of the angle formed by projecting a line of sight coincident to the intersecting point of the vertical and horizontal target line image shall not exceed the mil value specified in 3.7.1.1, as measured on the target. The reticle boresight point shall be coincident to the vertical target line image, as required in 3.7.1.1. Slip the knob scales to aline the four mil graduation lines to their fixed indexes.

4.5.7.1.2 Missile reticle position.- Project the missile reticle pattern into the optical path of the telescope by means of the reticle selector switch. Position the central cross of the reticle as in 4.5.7.1. Observe the position of the boresight collimating circle in relation to the image of the vertical line of the target reticle to determine conformance with the requirements of 3.7.1.2.

4.5.7.2 Missile reticle movement.- Project the missile reticle pattern in the field of view by means of the reticle selector switch. Move the line of sight through the central cross of the reticle by means of the respective knobs to determine conformance with the requirements specified in 3.7.2 from the position established in 4.5.7.1 as measured on the target.

4.5.7.3 Conventional reticle movement.- Project the conventional reticle pattern in the field of view by means of the reticle selector switch. From the position established in 4.5.7.1.1 move the line of sight by means of the respective knobs to determine conformance with the requirements specified in 3.7.2.

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4.5.7.4 Backlash in the missile reticle boresight mechanism.- Project the missile reticle pattern into the optical path of the telescope. With backlash excluded, rotate the horizontal boresight knob in a continuous clockwise direction until the vertical reticle line is in coincidence with the vertical target line. A mark shall be made on the dial control in true alignment with a mark on the housing which shall serve as a reference point. Proceed in the same clockwise direction for at least one half turn of the knob, then reverse rotation of the knob manually in a continuous counterclockwise direction (with backlash included) until the reference point is reached without overtravel. Observe the position of the vertical reticle line with reference to the vertical target line and read the mil scale of the target. Any difference between the vertical reticle line and the vertical target line is the amount of backlash. Repeat this operation with the vertical boresight knob, using the horizontal line of the scaled target to measure backlash. Backlash for each missile reticle boresight mechanism shall not exceed the limit specified in 3.7.4.

4.5.7.5 Backlash of the conventional reticle boresight mechanism.- Project the conventional reticle pattern into the optical path of the telescope. Rotate the deflection boresight knob manually in a continuous clockwise direction, with backlash excluded, until the reticle line is in coincidence with its respective mark on the target. Slip the knob scale to zero position. Proceed to rotate the knob in the same clockwise direction for at least one-half turn, then reverse rotation of the knob (with backlash included) until the line of sight is in coincidence with the recorded mark on target, without overtravel, and read the mil scale on the knob. Any difference between the readings as measured on the knob scale is the amount of backlash. Repeat this operation with the elevation boresight knob using the knob scale to record the readings. Backlash for each mechanism shall not exceed the limit specified in 3.7.4.

NOTE: During these operations (4.5.7.4 and 4.5.7.5), if overtravel occurs, the test shall be nullified and the procedures specified for backlash shall again be performed.

4.5.7.6 Reticle wander.- Articulate the telescope eyepiece housing assembly to the position specified in 3.6. Affix an auxiliary telescope to the eyepiece of the telescope under test. The auxiliary telescope shall contain a vertical and horizontal reticle line, and shall contain the etched data capable of measuring 1° reticle wander. The auxiliary telescope shall be of at least 3 power (3X). Obtain coincidence of the intersecting point of the auxiliary telescope reticle lines with the boresight point (intersection of the vertical and horizontal reticle lines) of the missile reticle, and plumb the vertical reticle line of the auxiliary telescope. The image of the telescope reticle boresight point shall not deviate in excess of 1° from the intersecting point of the auxiliary telescope reticle, when the articulating telescope is actuated as specified in 3.8. Reticle wander shall be measured on the auxiliary telescope reticle with reference to the telescope reticle.

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4.5.7.7 Reticle lean.- The test shall be performed with the aid of the auxiliary telescope positioned as specified in 4.5.7.6. The auxiliary telescope reticle shall be capable of measuring 1° reticle lean. Adjust the auxiliary telescope vertical line coincident to the vertical reticle line of the telescope missile reticle. Elevate and depress the telescope eyepiece housing as specified in 3.9. The image of the articulated telescope's vertical reticle line shall remain parallel to the auxiliary telescope's vertical reticle line within the tolerance specified in 3.9 during this actuation. Reticle lean shall be measured on the auxiliary telescope reticle with reference to the telescope's vertical reticle line.

4.5.7.8 Image tilt.- The test shall be performed with the aid of the auxiliary telescope positioned as specified in 4.5.7.6. Adjust the auxiliary telescope vertical reticle line plumb. Obtain coincidence of the intersecting point of the auxiliary telescope reticle lines with the intersecting point of the target vertical and horizontal lines. The image of the vertical target line shall be vertical within the tolerance specified in 3.10. Image tilt shall be measured on the target with reference to the vertical target line and the auxiliary telescope vertical reticle line.

4.5.7.9 Resolution.- This test shall be performed utilizing an auxiliary telescope with a magnification of at least 3 power (3X), a resolving power chart in a collimator projector representing the angular subtense for the seconds of arc specified for resolution, and a supporting medium to receive this test equipment as well as the telescope to be tested. The test method and procedure shall be in compliance with the "Resolution test" specified in MIL-O-13830, and the resolving power chart shall contain four sets of lines as shown in Figure 1 therein. The limit of resolution shall be within the value specified in 3.11 as determined by viewing through the auxiliary telescope.

4.5.7.10 Eyepiece focus.- The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X), and a diopter measuring range of at least plus and minus 4 diopters. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle, and set the dioptometer diopter scale to zero. Set the diopter scale of the telescope under test to zero diopter. Position the dioptometer behind the eyepiece assembly. Obtain a clear, sharp image of the telescope reticle by adjustment of the dioptometer diopter scale. The telescope reticle shall be in sharp focus within the limit specified in 3.12, as measured on the dioptometer diopter scale. Set the diopter scale of the articulating telescope to plus 4 diopter, and obtain sharp focus of the telescope reticle at the center of the field by adjustment of the dioptometer diopter scale. The focus of the telescope shall be plus 4 diopters as measured with the dioptometer. Repeat this test with the diopter scale of the telescope set to minus 4 diopters. Eyepiece focus of the telescope shall conform to the requirements of 3.12.

4.5.7.11 Reticle accuracy.- This test shall be conducted in accordance with MIL-O-13830. The angular subtense of the conventional reticle pattern between the zero meter range and the maximum meter etched data as measured on the target, shall not exceed the tolerance specified in 3.13.

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4.5.7.12 Parallax.- The test shall be performed with the diopter knob of the telescope set for best focus of the telescope reticle. The line of sight through each reticle in the telescope shall be directed to calibrated targets located at 600 meters, 1200 meters and infinity. Parallax is any apparent movement of the reticle lines in relation to the target image when the observer's head is moved from side to side or up and down. Parallax at the center of the field of view for each reticle shall be recorded. Total parallax between the reticles in the telescope shall not exceed the tolerance specified in 3.14.

4.5.7.13 Parallelism of missile reticle and image.- Sight through the eyepiece of the telescope and obtain coincidence of one end of the telescope's vertical reticle line with that of the vertical target line. Then observe coincidence at the opposite end of the vertical line. The vertical reticle line shall be parallel to that target line image within the tolerance specified in 3.15.1, as measured on the target. Deviation shall be noted and recorded.

4.5.7.14 Parallelism of conventional reticle and image.- Project the conventional reticle pattern into the optical system of the telescope by means of the reticle selector switch. Sight through the eyepiece of the telescope and obtain coincidence of one end of the telescope's vertical reticle dash line with that of the vertical target line. Then observe coincidence at the opposite end of the reticle lines. The vertical reticle dash line shall not deviate from the image of the vertical target line in excess of the tolerance specified in 3.15.2. Deviation shall be noted and recorded.

4.5.7.15 Variation of settings.- The angular difference between the recorded deviation for the missile reticle and conventional reticle obtained in 4.5.7.13 and 4.5.7.14 shall not exceed the tolerance specified in 3.15.3.

4.5.8 Diopter adjustment torque.- The test shall be performed with the aid of testing equipment in compliance with 4.4.2.3.3. The torque wrench shall be rotated in a clockwise direction and then in a counterclockwise direction from the diopter assembly zero scale setting to its appropriate stop. Readings shall be taken when the diopter adjustment assembly is in continuous motion. The running torque required to rotate the diopter adjustment assembly shall be within the values specified in 3.16.1 for the temperature indicated.

4.5.9 Filter transfer lever.- When the telescope is thermally stabilized at -40° and +150°F the filter transfer lever shall be subjected to a tactile check. The filter transfer lever shall conform to the requirements of 3.16.2 at each extreme temperature.

4.5.10 Articulation, torque.- Testing equipment in compliance with 4.4.2.3.3 shall be utilized for this test. An appropriate adapter shall be used with the torque measuring equipment to accommodate the objective tube assembly at the articulated joint. Position the telescope as specified in 3.16.3. Torque shall be measured while rotating the objective tube through its full movement about the articulated joint with the eyepiece housing assembly immobilized. The maximum running torque required to rotate the objective tube shall not exceed the value specified in 3.16.3 for the temperature indicated.

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4.5.11 Disengagement of boresight knob locks.- Testing equipment used to measure direct force, push or pull, in pounds with a minimum range of zero to ten pounds, the accuracy of which shall be plus or minus two (2) ounces throughout the entire zone of operation, shall be utilized. This testing equipment shall be equipped with suitable adapters to accommodate the testing of the boresight knobs. The test shall be conducted by use of the above equipment. The force (pull) required to disengage the knobs from their lock position shall meet the value specified in 3.16.4 for the temperature indicated.

4.5.12 Rotation of boresight knobs.- Testing equipment in compliance with 4.4.2.3.3 shall be utilized for this test. The appropriate adapter shall be used with the torque measuring equipment to accommodate the boresight knobs. The maximum running torque required to rotate each knob throughout its range of movement in its unlocked position shall not exceed the value specified in 3.6.14.1 for the temperatures indicated.

4.5.13 Lamp housing assemblies.- The lamp housing assemblies shall be subjected to a tactile examination. The missile and conventional reticle lamp housing assemblies shall conform to the requirements of 3.17 at the temperatures indicated.

4.5.14 Reticle illumination.- The illuminated reticle etchings shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the brightness of a reticle line, the defect shall be cause for rejection. Opaque areas shall be free of pinpoint holes, cracks or chips.

4.5.15 Circuitry.- A continuity check of the telescope circuits shall be made using standard measuring equipment to determine conformance with the requirements of 3.18.2.

4.5.16 Dielectric strength.- The telescope shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment to determine conformance with the requirements of 3.18.3. All lamps shall be removed for this test.

4.5.17 Insulation resistance.- Immediately after satisfactorily withstanding the dielectric strength test, the telescope shall be subjected to the insulation resistance test in accordance with Method 302, Test Condition B of MIL-STD-202. Measurements between insulating points and ground shall be made utilizing standard measuring equipment to determine conformance with the requirements of 3.18.4. All lamps and ground connection shall be removed for this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8624500. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The Telescope XM119 is primarily intended as the direct fire telescope for the XM551 Sheridan Vehicle.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1240-A431Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. *IS THE SPECIFICATION RESTRICTIVE?*

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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